MODULE 4

4.1 MODERN COMMUNICATION SYSTEM SCHEME

- **Communication** is the science and practice of transmitting information from one point to another.
- **Communication engineering** also deals with the techniques of transmitting information. i.e; electrical communication, in which information is transmitted through electrical signals.
- Thus, **electrical communication** is a process by which the information/message is transmitted from one point to another, from one person to another, or from place to another in the form of electrical signals, through some communication link.
- A **basic communication system** provides a link between the information source and its destination. The process of electrical communication involves sending, receiving, and processing information in electrical form. A basic communication system consists of certain units, called constituents, subsystems, or stages.
- The information to be transmitted passes through a number of stages of the communication system prior it reaches its destination.

Figure 1.1 shows a **block schematic diagram of the most general form of basic communication system.**

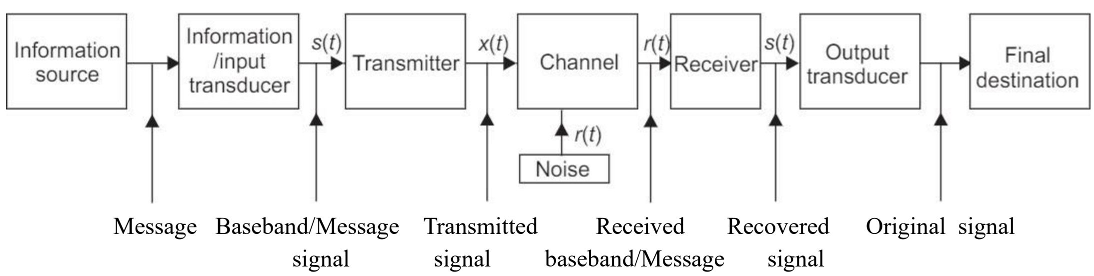

Fig. 1.1 Schematic block diagram of a basic communication system in most general form

- The **main constituents (stages) of basic communication system** are:

(i) Information source and input transducer
(ii) Transmitter
(iii) Channel or medium
(iv) Noise
(v) Receiver
(vi) Output transducer and final destination.

- There are many **types of communication systems,** e.g. analog, digital, radio, and line communication systems. The different communication systems apply different principles of operation and physical appearance to each constituents, in accordance with its type.

➢ Information Source and Input Transducer

- **A communication system transmits information from an information source to a destination and hence the first stage of a communication system is the information source. The physical form of information is represented by a message that is originated by an information source,** e.g. a sentence or paragraph spoken by a person is a message that contains some information. The person, in this case, acts as information source. Few other familiar **examples of messages are voice, live scenes, music, written text, and e-mail.**

- A communication system transmits information in the form of electrical signal or signals**. If the information produced by the source is not in an electrical form, a transducer, to convert the information into electrical form.**

A transducer is a device that converts a non-electrical energy into its corresponding electrical energy called signal and vice versa, e.g. during a telephone conversation, the words spoken by a person are in the form of sound energy. This has to be converted to its equivalent electrical form prior it is transmitted.

- An example of a transducer is a microphone. Microphone converts sound signals into the corresponding electrical signals. Similarly, a television (TV) picture tube converts electrical signals into its corresponding pictures. Some other **examples of transducers are movie cameras, Video Cassette, Recorder (VCR) heads, tape recorder heads, and loudspeakers.**

- The information produced by the information source is applied to the next stage, termed the information or Function input transducer. This in turn, produces an electrical signal value corresponding to the information as output. This electrical signal is called **the baseband signal. It is also called a message signal, an information signal, an intelligent signal, or an envelope. In the communication theory, the (a) baseband signal is usually designated by s(t).**

- **There are two types of signals**. (a) analog signal, and (b) digital signal.

(a) Analog Signal

- **An analog signal is a function of time, and has a continuous range of values.** There is a definite function value of the analog signal at each point of time.

An example of analog signal or analog wave form is a pure sine wave form. A practical **example** of an analog signal is a **voice signal**.

- When a voice signal is converted to electrical form by a microphone, we get a corresponding electrical analog signal. One can see this electrical signal on an oscilloscope. An analog signal, this wave form has definite values at all points of time.

- Analog signals are shown in Fig. 1.2. (b)

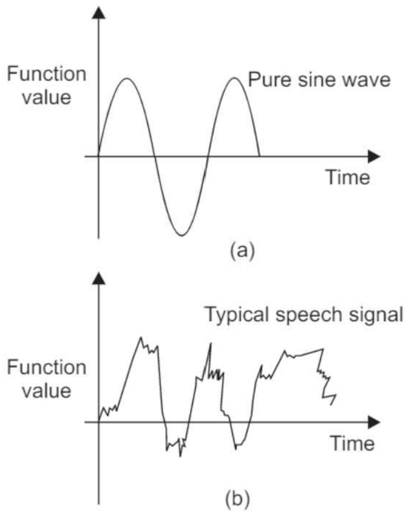

Fig. 1.2 Analog signals: (*a*) Pure sine wave, (*b*) Typical speech signal

(b) Digital Signal

- **A digital signal does not have continuous function values on a time scale. It is discrete in nature, i.e., it has some values at discrete timings.** In between two consecutive values, the signal values is either zero, or different value. A familiar **example** of a digital signal is **the sound signal produced by drumbeats.**
- Figure 1.3 shows a graphical representational of a digital signal.

Digital signals in their true sense correspond to a binary digital signal, where the discrete amplitude of the signal is coded into binary digits represented by '0' and '1 '.

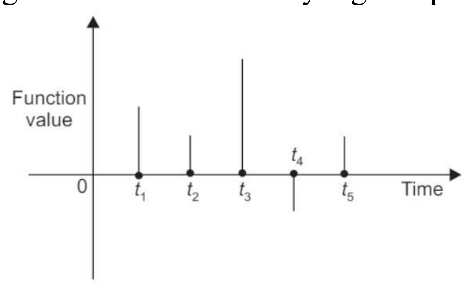

Fig. 1.3 Digital signal

- The analog signal, which is continuous in time, is converted to discrete time, using a procedure calling **sampling**.
- The continuous amplitude of the analog signal is converted to discrete amplitude using a process called **quantization**.
- Sampling and quantization are together termed as **analog-to-digital conversion (ADC)** and the circuitary that performs this operation is called an analog to-digital converter.

➢ Transmitter

- The next block in the communication system of Fig. 1.1 is the transmitter. The base band signal, which is the output of an input transducer, is input to the transmitter. This base band signal, s(t), is suitable for transmission in the form in which it is generated by the transducer.
- The transmitter section processes the signal prior transmission. The nature of processing depends on the type of communication system. However, the processing carried out for signal transmission in the analog form is different from signal transmission in the digital form**.**

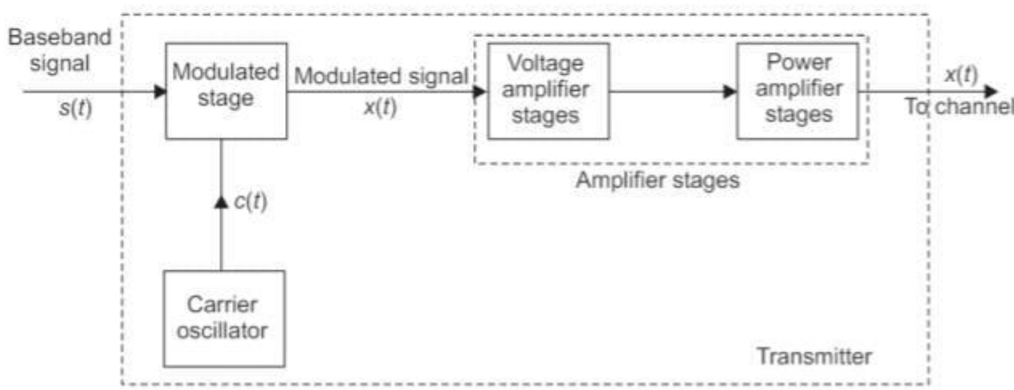

Fig. 1.5 Block diagram representing schematic of an analog transmitter section

Fig. 1.5 Block diagram representing schematic of an analog transmitter section

There are two following options for processing signals prior transmission:

(i) The baseband signal is transmitted without translating it to a higher frequency spectrum. Here we call the system as **a baseband communication** system, because the baseband signal is transmitted without translating it to a higher frequency spectrum. However, some processing of the signal is required prior its transmission, e.g. a train of pulses that are to be transmitted can be replaced by a series of two sine waves of different frequencies prior to transmission. One of these two frequencies represent a low and the other represents a high value of the digital pulse.

Therefore, **the baseband signal is converted into a corresponding series of sine waves of two different frequencies prior to transmission.** Figure 1.4 illustrates this processing.

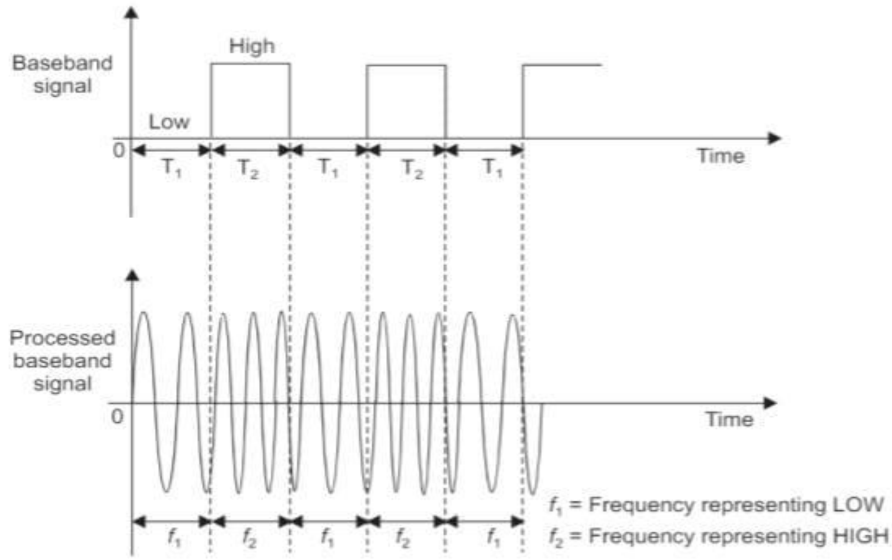

Fig. 1.4 The processing of a baseband signal

(ii) The baseband signal, which lies in the low frequency spectrum, is translated to a higher frequency spectrum. Here we call the communication system as a **carrier communication** system. In this system**, the baseband signal is carried by a higher frequency signal, called the carrier signal.** The process of translating a low frequency baseband signal to higher frequency spectrum using a high frequency carrier signal is called **Modulation.**

- Figure 1.5 shows the baseband signal, s(t) applied to the modulated stage. This stage translates the baseband signal from its low frequency spectrum to high frequency spectrum. This stage also receives another input called **the carrier signal**, c(t), which is generated by a high frequency carrier oscillator.

- Modulation takes place at this stage with the baseband and the carrier signals as two inputs after modulation, the baseband signal is translated to a high frequency spectrum and the carrier signal is said to be modulated by the baseband signal. The output of the modulated stage is called the **modulated signal,** and is designated as x(t).

- Now, if the baseband signal is a digital signal, the carrier communication system is called a **digital communication system**.

- The digital modulation methods are employed for this. If the baseband signal is an analog signal, the carrier communication system is called as an **analog communication system** and for processing the analog modulation techniques are used.

- The voltage of the modulated signal is then amplified to drive the last stage of the transmitter, called the power amplifier stage (Fig. 1.5). This stage amplifies the power of the modulated signal and thus it carries enough power to reach the receiver stage of the communication system. Finally, the signal is passed to the transmission medium or channel.

➢ Channel Or Medium

After the required processing, the transmitter section passes the signal to the transmission medium. The signal propagates through the transmission medium and is received at the other side by the receiver section. **The transmission medium between the transmitter and the receiver is called a channel.**

The channel is a very important part of a communication system as its characteristics add many constraints to the design of the communication system, e.g. most of the noise is added to the signal during its transmission through the channel. The transmitted signal should have adequate power to withstand the channel noise. Further, the channel characteristics also impose constraints on the bandwidth. **The bandwidth is the frequency range that can be transmitted by a communication system**. Moreover, the channel characteristics are also taken into consideration as a design parameter while designing the transmitting and receiving equipment.

The transmitting power, signal bandwidth, design and cost of the communication system are affected by channel characteristics.
Depending on the physical implementations, one can classify the channels in the following two groups:

➢ <u>Hardware Channels</u>

➢ **These channels are manmade structure which can be used as transmission medium. There are following three possible implementations of the hardware channels.**

- **Transmission lines**
- **Waveguides**
- **Optical Fiber Cables (OFC)**

The examples of transmission lines are twisted-pair cables used in landline telephony and coaxial cables used for cable TV transmission.
However, transmission lines are not suitable for ultra high frequency (UHF) transmission. To transmit signals at UHF range, waveguides are employed as medium. Waveguides are hollow, circular, or rectangular metallic structures. The signals enter the waveguide, are reflected at the metallic walls, and propagate towards the other end of the waveguide.
Optical fiber cables are highly sophisticated transmission media, in the form of extremely thin circular pipes. Signals are transmitted in the form of light energy in optical fiber cables.

- In general, **there is always a physical link between the transmitter and receiver in hardware channels.** A communication system that makes use of a hardware channel is called as a **line communication system**, e.g. landline telephony and cable TV network.

➢ Software Channels

- **There are certain natural resources which can be used as the transmission medium for signals. Such transmission media are called software channels.** The possible natural resources that can be used as **software channels are: air or open space and sea water.**
- **In communication systems that use software channels there is no physical link between the transmitter and the receiver**. The transmitter passes the signals in the required form to the software channel. The signals propagate through the natural resource and reach the receiver.
- The most widely used software channel is air or open space. The signals are transmitted in the form of electromagnetic (em) waves', also called radio waves. Radio waves travel through open space at a speed equal to that of light ($c = 3 \times 10^8$ m/s).
- The transmitter section converts the electrical signal into em waves or radiation by using a transmitting antenna. These waves are radiated into the open space by the transmitting antenna.
- At the receiver side, another antenna, called the receiving antenna, is used to pick up these radio waves and convert them into corresponding electrical signals. Systems that use radio waves to transmit signals through open space are called **radio communication systems**, e.g. radio broad cast, television transmission, satellite communication, and cellular mobile communication.

- Radio signals are transmitted through electromagnetic(em) waves, also referred as **radio waves**, in a radio communication system. The radio waves have a wide frequency range starting from a few ten kilo Hertz (Hz) to several thousand Mega Hertz (MHz). This wide range of frequencies is referred as **the radio frequency (RF) spectrum**.

The RF spectrum is classified according to the applications of the spectrum in different service areas.

Table 1.1 shows the **classification of the RF spectrum** along with the associated applications in communication systems.

Table 1.1**: Classification of radio frequency (RF) spectrum along with the associated applications in communication systems**.

Radio frequency range	Wavelength (meters)	Class	Applications
10-30 kHz 30-300 kHz 300-3000 kHz 3-30 MHz 30-300 MHz 300-3000 MHz 3000-30,000 MHz	$3 \times 10^4 - 10^4$ — 103 102 - 10 10 - 1.0 1.0 - 0.1 0.1 - 0.01	Very Low Frequency (VLF) Low Frequency (LF) Medium Frequency (MF) High Frequency (HF) Very High Frequency (VHF) Ultra High Frequency (UHF) Super High Frequency (SHE)	Point-to-point communication (long distance) Point-to-point communication (long distance) and navigation Radio broadcasting Overseas radio broadcasting, Point-to-point radio telegraphy, and telephony FM broadcast, television, and radar Television and navigation Radar navigation and radio relays

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➢Noise

- In electronics and communication engineering**, noise is defined as unwanted electrical energy of random and unpredictable nature present in the system due to any cause. Noise is an electrical disturbance, which does not contain any useful information. Thus, noise is a highly undesirable part of a communication system, and have to be minimized.** We may note that noise cannot be eliminated once it is mixed with the signal. When noise is mixed with the transmitted signal, it rides over it and deteriorates it waveform. This results in the alteration of the original information so that wrong information is received.
- The receiver processes the signal to cover the original information produced by the information source at the transmitter end. If the amplitude of the noise is comparable with that of the signal, then the noise may render the transmitted signal unintelligible, and the receiver recovers nothing but the noise.
- In order to avoid this undesirable situation, the system designer can make the signal adequately powerful prior to transmitting it. This enables the signal to withstand the noise. In fact, the system designer increases the power of the signal in comparison with that of the noise. This increases the ratio of the signal power to the noise power, i.e. **SNR (signal to noise ratio: The ratio of Signal power to the noise power).**
- The designer provides adequate signal strength at the time of transmission so that a high SNR is available at the receiver.
- The noise block in Fig. 1.1 represents the total noise present in the system, contributed by all the sources. The noise signal n(t), is applied to the channel block.

However, this does not mean that the noise is intermingled with the signal only during its propagation through the channel.

- In fact, the channel contributes the major part of the noise. However, other noise sources along the communication chain can also add noise to the signal.
- The noise may also be mixed with the signal from within the transmitting and receiving equipments. Since it is not possible to show all the individual sources of noise along the communication chain, we have shown only one noise block in Fig. 1.1, beneath the channel block, as the channel in the main source of noise.
- The noise introduced by the transmission medium is called **extraneous or external noise**.
- The main cause of the **internal noise** is the thermal agitation of atoms and electrons of electronic components used in the equipment.

❖ External Noise

- External noise is introduced by the transmission medium. External noise is so called b cause the transmission medium external to the receiver. One can divide the external noise into the following three groups:

(i) **Atmospheric or static noise**

The main source of this noise are lightning discharges and natural electrical disturbances in the atmosphere. These electrical disturbances are received by the receiving antenna and are mixed with the signal as noise. As atmospheric noise is caused by nature and hence there is no way to eradicate it. The only remedy is that the receiver is provided with ample gain so the atmospheric noise is countered.

(ii) **Man-made or industrial noise**

The main sources of this noise are automobile and aircraft ignition, electric motors, heavy electrical machines, switching gears, leakage from high-voltage electrical supply lines, and fluorescent lights. This noise is heavier in industrial areas than in non-industrial areas, and this is why it is called as industrial noise. The transmitting equipment should be installed away from the industrial areas to avoid this type of noise.

(iii) **Extraterrestrial or Space noise**

Outer space is also a source of external noise that comes from sun and stars. This noise is of two types: (a) solar noise, and (b) cosmic noise.

The source of solar noise is the sun, which continuously emits electrical energy which gets mixed with the transmitting radio signal and appears as noise.

The source of cosmic noise is also stars, which also radiate electrical energy over a wide frequency range covering the radio frequency spectrum.

Galaxies are also the source of space noise.

❖ <u>Internal noise</u>

The noise introduced within the receiver is known as internal noise. This noise can be grouped into: (a) Thermal or Johnson's noise or white noise, and (b) Shot or transistor noise.

a) Thermal or Johnson's noise or white noise

This noise is generated with resistors used in the circuit. This noise appears due to the random motion of electrons and molecules inside the resistor. The resistive part of the complex impedance in a circuit is also a source of thermal noise.

b) Shot or Transistor noise

This noise is generated inside a transistor used in an amplifier. Shot noise appears because of the rando movement of electron and holes inside the transistor and the presence of random electrons a he output termina of the transistor in an amplifier.

❖ <u>SNR and Noise Figure (F)</u>

The SNR as the ratio of the signal power to the noise power at a point in the circuit. SNR is the measure of the signal power relative to the noise power at a particular point in a circuit.

Now, if Ps is signal power and Pn is noise power, then SNR expressed as S/ N, is given as $\frac{S}{N} = \mathbf{P_s / P_n = (V_s)^2 R / (V_n)^2 R = (V_s)^2 / (V_n)^2}$ **Signal to noise ratio expressed in decibel is given by (S/N) in dB =** $\mathbf{10 \log [(V_s)^2 / (V_n)^2]}$

- **Noise Factor is defined as the ratio of SNR at the input to the SNR at the output' Noise figure is simply the noise factor expressed in decibels.**
- **N = 10 log10 (Si/Ni) / (So/No) where Si is the signal at the input, Ni is the noise at the input, So is the signal at the output, No is the noise at the output.**

➢ Receiver

- **The task of the receiver is to provide the original information to the user. This information is altered due to the processing at the transmitter side. The signal received by the receiver, thus, does not contain information in its original form. The receiver system receives the transmitted signal and performs some processing on it to recover the original baseband signal.**
- **The signal received by the receiver by r(t) in Fig. 1.1. This signal contains both the transmitted signal, x(t), and the noise, n(t), added to it during transmission. The function of the receiver section is to separate the noise from the received signal, and then recover the original baseband signal by performing some processing on it. The receiver receives a weak signal because the transmitted signal loses its strength during its propagation through the channel.** This occurs due to the attenuation of the signal.
- A voltage amplifier first amplifies the received signal so that it becomes strong enough for further processing, and then recovers the original information. The original baseband signal is recovered by performing an operation opposite to the one performed by the transmitter section.
- The detailed block diagram of a typical receiver section is shown in Fig. 1.7.

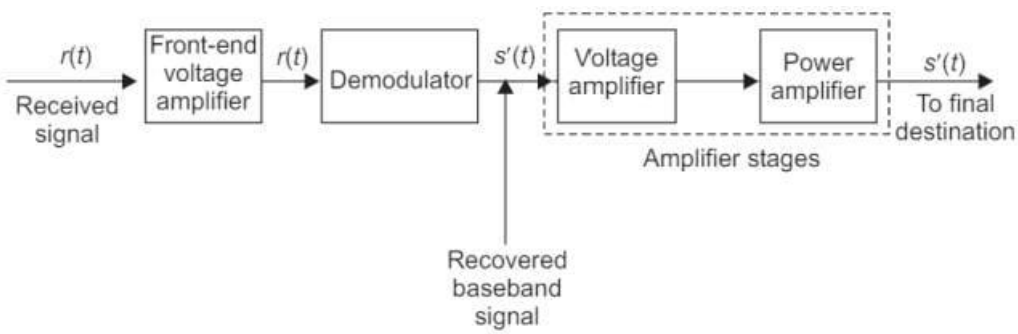


Fig. 1.7 Detailed block diagram of a typical receive section

- The transmitter performs modulation on the baseband signal to translate it to a higher spectrum from its low frequency spectrum. The receiver, in turn, performs an operation known as demodulation, which brings the baseband signal from the higher frequency spectrum to its original low-frequency spectrum.
- The **demodulation** process removes the high frequency carrier from the received signal and retrieves the original baseband. This occurs in a carrier communication system. The baseband communication systems, assume that the transmitter replaces the digital baseband signal with a series of two sinusoidal wave forms of different frequencies as shown in Fig. 1.4.
- When the receiver receives this signal, it recovers the original baseband pulse by replacing the two sinusoidal waveforms with the corresponding original levels.

- The recovered baseband signal is then handed over to the final destination, which uses a transducer to convert this electrical signal to its original form.
- It is essential that enough signal power is given to the transducer so that it satisfactorily reproduces the message. Therefore, prior to handing over the recovered baseband signal to its final destination, the voltage and power are amplified by the amplifier stages.
- From Fig. 1.7 it is evident that the received signal, r(t), is first amplified by the front-end voltage amplifier. This is done to strengthen the received signal, which is weak and to facilitate easy processing.
- Next, this signal is given to the demodulator, which in turn, demodulates the received signal to recover the original baseband signal. Interestingly, the type of demodulation is based on the type of modulation employed at the transmitter.
- After recovering the original baseband signal, its voltage and power is amplified prior it to final destination block.

➢ Final Destination

- **The final destination is the last block of the communication chain. The destination is the user that receives the information and interprets it for useful purposes.**
- The user can either be a human being or a machine that can make certain decisions based on the information received, e.g. if a message is transmitted in the form of spoken words, then the user can be a human being who makes certain decisions on its reception. A live telecast of a football match is meant a human being as a user. In meant for a human being as a user. In some cases, one can use a computer for the interpretation of the received information and takes appropriate action based on the information.
- **The destination block converts the baseband signal, which contains information in electrical form back into its original non-electrical rm back into its original non-electrical form, to be delivered to the user. The constituents of the destination block are an output transducer, which converts the baseband signals from its electrical form to its original non-electrical form and the user**.

4.2 MULTIPLEXING

- This is a technique that is most widely used in nearly all types of communication systems, radio and line communication systems. Basically, **multiplexing is a process which allows more than one signal to transmit through a single channel. Clearly, multiplexing facilitates the simultaneous transmission of multiple messages over a single transmission channel.**

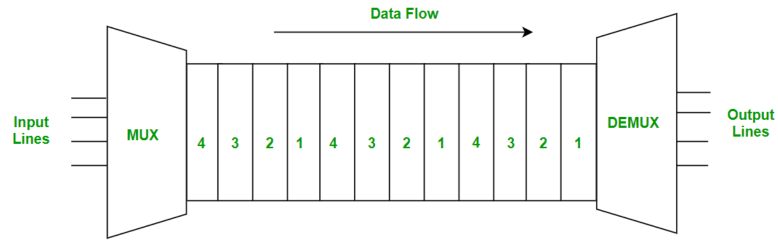
- **Multiplexing allows the maximum possible utilization of the available bandwidth of the system. Bandwidth is an important entity in any communication system**.
- The use of **multiplexing also makes the communication system economical** because more than one signal can be transmitted through a single channel.
- **Multiplexing is possible in communication system only through modulation.**

- To consider multiplexing, let us consider the following example. If many people speak loudly and simultaneously, then it becomes nearly impossible to understand their conversion because the overall result is noise. This noise is the result of mixing of all the speeches. The human ear is not capable of separating these intermingled speeches and therefore no intelligent words are communicated to brain.
- The same situation is now applied to the transmission of audio signals. These audio signals may come from, say ten different persons. **While the speech frequency of different persons will be different, all the ten signals will lie in the same audio range of 20 Hz to 20 kHz. If all these baseband audio signals are simultaneously transmitted through a single channel, then they will be mixed together. The transmitter will transmit these mixed signals, and the receiver will receive them. The purpose of the receiver is to deliver the audio signals in their original form. If all the received signals lie within the same audio range, the receiver is not capable of separating them into individual signals**, similar to the case with human ears.
- In order to avoid this difficulty, each signal can be translated to a different frequency spectrum, such that every signal differs in its transmitted frequency through modulation. Therefore, if all the baseband signals are modulated, i.e., translated to higher frequencies by using different carrier frequencies, then each signal is easily distinguishable from the other although they all lie within the same audio band. At the transmitter they can be mixed and transmitted.
- At the receiver, the different signals can be easily separated because they are at different frequencies, and these can be delivered to the next stages of the receiver for further processing. M**ultiplexing becomes possible only because of modulation.**

The important types of multiplexing are:

- **Time Division Multiplexing**

In TDM, the channel is divided into several time slots, and each signal is transmitted during its allocated time slot. As a result, several signals share the channel without interfering with each other. Eg multiple users sharing the printer. The printer processes the print jobs one at a time, in the order they were received, during their allocated time slots. Another eg: Traffic lights at an intersection use time division multiplexing to manage the flow of traffic.



- **Frequency Division Multiplexing**
- **Divides bandwidth into subchannels of different frequencies, each channel carrying a signal in parallel. Guard bands will be provided between the channels to avoid mixing up of channels.**

  - **Examples:** Radio and television broadcasting, where multiple radio signals at different frequencies are transmitted simultaneously.
  - Cable television, which carries many television channels on a single cable.
  - Commercial broadcast radio (AM and FM radio), where each station gets its own frequency band.